Understanding Financial Markets through Econometrics and Machine Learning: Evidence from U.S. Equity Data

Author Contribution Statement

This paper was developed as part of a group project in the Master of Science in Financial Engineering program at WorldQuant University.

Rasheed Uthman: Led the preparation of the manuscript, organized the overall structure of the paper, wrote the introduction, literature review, methodology, empirical results, discussion, and conclusion, integrated the econometric and machine learning analyses into a unified research framework, and prepared the final version of the manuscript.

Anas Ebaid: Contributed to the project through data preparation, empirical analysis, programming support, and technical discussions.

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Abstract

Forecasting financial markets remains a central challenge in empirical finance because asset prices respond continuously to changing economic conditions, investor expectations, and market uncertainty. During the period from 2019 to 2023, U.S. financial markets experienced unprecedented disruptions, including the COVID-19 pandemic, the subsequent economic recovery, and a rapid tightening of monetary policy. These events underscore the need for analytical frameworks that not only improve prediction but also explain the statistical behaviour of financial data.

In this study, I examine whether combining classical econometric methods with modern machine learning techniques provides a more informative approach to analysing financial markets. I first investigate the time-series properties of U.S. macroeconomic variables using the Augmented Dickey–Fuller test, Johansen cointegration analysis, Error Correction Models, and a Markov Regime Switching Model. I then construct financial features from daily observations of ten large U.S. companies and the S&P 500 Index between 2019 and 2023. To address the high correlation among engineered features, I apply Principal Component Analysis before estimating a Ridge Regression model to predict market returns.

The empirical analysis shows that the macroeconomic variables are non-stationary in levels but become stationary after first differencing, supporting the use of cointegration and error correction techniques. The financial analysis further demonstrates that feature engineering and dimensionality reduction provide a compact representation of market information for prediction. Overall, this study illustrates how econometric analysis and machine learning can be integrated within a single empirical framework to better understand financial markets while maintaining both statistical interpretation and predictive capability.

Introduction

1.1 Background

Between 2019 and 2023, U.S. financial markets experienced one of the most volatile periods in recent history. In early 2020, the COVID-19 pandemic triggered a sharp decline in global equity markets as governments imposed lockdowns and economic activity slowed dramatically. Within months, unprecedented fiscal stimulus and accommodative monetary policy supported a rapid recovery in asset prices. By 2022, however, rising inflation prompted the U.S. Federal Reserve to begin one of its most aggressive cycles of interest-rate increases in decades. These events fundamentally changed investor expectations and highlighted how quickly financial markets can respond to economic shocks.

For investors, policymakers, and researchers, these episodes reinforced an important challenge: financial markets generate enormous amounts of information, yet identifying which information is useful for explaining or predicting market movements remains difficult. Stock prices are influenced not only by firm-specific news but also by macroeconomic conditions, monetary policy, and changes in market sentiment. As these factors evolve over time, relationships observed during one period may weaken or even disappear under different economic conditions.

These challenges suggest that accurate financial analysis requires more than prediction alone. Before developing forecasting models, it is important to understand the statistical properties of the underlying data. Financial and macroeconomic variables often exhibit non-stationarity, structural changes, and strong correlations, all of which can lead to misleading conclusions if ignored. Econometric methods such as the Augmented Dickey–Fuller test, Johansen cointegration analysis, and Error Correction Models provide a systematic framework for identifying these characteristics and distinguishing short-run fluctuations from long-run economic relationships.

At the same time, advances in computing and the increasing availability of financial data have encouraged the adoption of machine learning techniques in quantitative finance. Rather than replacing econometric analysis, machine learning provides practical tools for handling high-dimensional datasets, constructing informative features, and improving predictive performance. Techniques such as Principal Component Analysis and regularized regression are now widely used in portfolio management, risk modelling, and algorithmic trading because they can extract useful information from complex financial data while reducing problems associated with multicollinearity.

Motivated by these developments, this study investigates whether econometric methods and machine learning can be combined within a single analytical framework to better understand financial markets. Using U.S. macroeconomic data together with daily equity data for ten large U.S. companies and the S&P 500 Index between 2019 and 2023, I first examine the statistical properties of the data before applying feature engineering, dimensionality reduction, and predictive modelling. Rather than asking whether one approach is superior to the other, I explore how the strengths of both approaches can be integrated to produce a more reliable and interpretable framework for financial market analysis.

1.2 Research Problem

Financial market prediction has traditionally followed two separate paths. Econometric research focuses on explaining the statistical and economic relationships among financial variables, while machine learning research primarily emphasizes predictive performance. Although both approaches have advanced significantly, they are often applied independently, limiting our understanding of how statistical inference and prediction can reinforce one another.

This separation presents an important challenge because financial data are inherently complex. Asset prices and macroeconomic variables frequently exhibit non-stationarity, structural breaks, and strong correlations among explanatory variables. Ignoring these characteristics may lead to unreliable statistical inference, unstable prediction models, or unnecessary model complexity. At the same time, focusing solely on econometric analysis

may overlook the ability of modern machine learning techniques to extract useful information from high-dimensional financial data.

Motivated by this gap, this study investigates whether a sequential framework that combines econometric analysis with machine learning can provide a more robust approach to financial market analysis. Specifically, I first examine the statistical properties of the data using unit root tests, Johansen cointegration analysis, Error Correction Models, and a Markov Regime Switching Model. I then apply feature engineering, Principal Component Analysis,

and Ridge Regression to evaluate whether reducing feature dimensionality improves the prediction of U.S. equity market returns. By integrating these methods within a single empirical framework, this study seeks to bridge the gap between economic interpretation and predictive modelling.

1.3 Research Objectives

The primary objective of this study is to investigate whether integrating econometric methods with machine learning provides a more robust framework for analysing financial markets and predicting stock returns.

To achieve this objective, I pursue the following specific objectives:

I examine the statistical properties of selected U.S. macroeconomic variables before developing predictive models.
I investigate the long-run relationships and short-run dynamics among the macroeconomic variables.
I identify changes in market conditions using a Markov Regime Switching model.
I construct and transform financial features from daily U.S. equity data to improve market prediction.
I evaluate whether dimensionality reduction and Ridge Regression improve stock return prediction.
I assess whether integrating econometric analysis with machine learning provides a more informative framework for financial market analysis.

1.4 Research Questions

This study is guided by the following research questions:
Do the selected macroeconomic variables exhibit unit roots and long-run equilibrium relationships?
Can Error Correction Models and Markov Regime Switching Models adequately capture the dynamic behaviour of financial time series?
Does feature engineering improve the representation of information contained in daily U.S. equity prices?
Does Principal Component Analysis retain useful predictive information while reducing the dimensionality of financial data?
Does integrating econometric methods with machine learning improve financial market analysis and stock return prediction?

1.5 Contributions of the Study

This study makes several contributions to the growing literature at the intersection of financial econometrics and machine learning.
First, it presents an integrated analytical framework that combines classical econometric techniques with modern machine learning methods rather than treating them as separate approaches. By examining the statistical properties of financial and macroeconomic data before developing predictive models, the study demonstrates how economic interpretation and prediction can complement one another.
Second, the study illustrates the practical value of feature engineering in financial market analysis. Instead of relying only on raw price data, a set of technical indicators is constructed to capture information about market trends, momentum, and price variability. Principal Component Analysis is then used to summarize this information into a smaller number of features while preserving most of the variation contained in the original dataset.

Third, the empirical analysis provides a reproducible workflow that integrates data collection, econometric modelling, feature engineering, dimensionality reduction, and predictive modelling within a single framework. This approach reflects the type of workflow increasingly used in quantitative finance, financial engineering, and empirical asset pricing research.
Finally, the study demonstrates how econometric reasoning can strengthen machine learning applications in finance. Rather than viewing prediction as the only objective, it emphasises the importance of understanding the statistical structure of financial data before applying predictive algorithms. This perspective is particularly relevant for researchers and practitioners seeking models that are both interpretable and empirically robust.

. Literature Review and Theoretical Motivation

2.1 Financial Markets and the Challenge of Prediction

Predicting financial markets remains one of the most challenging problems in empirical finance. Asset prices respond continuously to changes in economic conditions, monetary policy, corporate performance, and investor expectations. As a result, financial time series often exhibit non-stationarity, structural changes, and periods of heightened volatility, making reliable prediction difficult.

These challenges became particularly evident during the 2019–2023 period examined in this study. The COVID-19 pandemic, rising inflation, and successive increases in U.S. interest rates significantly altered market conditions, creating an environment in which historical relationships were not always stable. Models estimated during one period may therefore perform poorly when economic conditions change.

Rather than focusing solely on prediction, I first seek to understand the statistical properties of the data. Establishing whether macroeconomic variables satisfy the assumptions required for time-series analysis provides a stronger foundation for subsequent modelling. I therefore begin my empirical analysis with time-series econometric methods before developing machine learning models for stock return prediction.

2.2 Time-Series Econometrics as the FoundationI begin the empirical analysis by examining whether the macroeconomic variables are stationary. Variables such as real GDP, real personal consumption expenditure, and the Federal Funds Rate often exhibit persistent trends. Estimating models with non-stationary data can produce misleading results, making stationarity testing an essential first step.
I apply the Augmented Dickey–Fuller (ADF) test, which is estimated as

$$\Delta y_t=\alpha+\beta t+\gamma y_{t-1}+\sum_{i=1}^{p}\delta_i\Delta y_{t-i}+\varepsilon_t,$$

where y_t is the observed time series, Δ denotes the first-difference operator, p is the lag length, and ε_t is the error term. The null hypothesis, $H_0:\gamma=0$, indicates that the series contains a unit root, while the alternative hypothesis, $H_1:\gamma<0$, implies stationarity.
After determining the order of integration, I proceed to examine whether the macroeconomic variables share a long-run equilibrium relationship through cointegration analysis.

2.3 Long-Run Equilibrium and Short-Run Adjustment

Once the variables are found to be integrated of the same order, I examine whether they share a long-run equilibrium relationship. Cointegration analysis is widely used in financial econometrics because many macroeconomic variables move together over time despite experiencing short-run fluctuations.
I use the Johansen cointegration approach, which represents the long-run relationship as

[
\Pi=\alpha\beta',
]
where (\beta) contains the cointegrating vectors and (\alpha) measures the speed at which variables adjust toward long-run equilibrium.
When cointegration exists, the short-run dynamics can be modelled using an Error Correction Model (ECM):
[
\Delta y_t=\alpha+\sum_{i=1}^{p}\Gamma_i\Delta y_{t-i}+\lambda EC_{t-1}+\varepsilon_t,
]
where (EC_{t-1}) is the lagged error-correction term and (\lambda) measures the speed of adjustment toward equilibrium after a short-run shock.
In this study, I use Johansen cointegration and the ECM to determine whether real GDP, real personal consumption expenditure, and the Federal Funds Rate maintain a stable long-run relationship before examining changes in market regimes.

2.4 Market Regimes and Structural Change
Financial markets do not operate under constant conditions. Economic shocks, policy changes, and financial crises can alter the relationship between macroeconomic variables and asset prices. Models that assume constant parameters may therefore fail to capture these structural changes.
To account for changing market conditions, I estimate a Markov Regime Switching model:
[
y_t=\mu_{S_t}+\beta_{S_t}X_t+\varepsilon_t,
]
where (S_t) denotes the unobserved market regime, (\mu_{S_t}) is the regime-specific intercept, and (\beta_{S_t}) represents the regime-dependent coefficients.
Applying this model allows me to identify shifts in economic conditions before moving from macroeconomic analysis to daily U.S. equity data and stock return prediction.

2.5 Linking Macroeconomic Conditions to Equity Markets
Macroeconomic conditions influence financial markets through interest rates, economic growth, inflation, and consumer spending. Changes in these variables affect corporate earnings, investment decisions, and investor expectations, ultimately influencing stock prices.
To examine these relationships, I complement the macroeconomic analysis with daily equity data for ten large U.S. companies and the S&P 500 Index from 2019 to 2023. Instead of focusing on a single industry, I intentionally select firms from technology, banking, healthcare, energy, retail, consumer goods, entertainment, and aerospace. This diversified sample reduces sector-specific bias and provides a broader representation of the U.S. equity market.
Having established the macroeconomic environment, I next transform the stock price data into informative financial features for predictive modelling.

2.6 Feature Engineering for Financial Prediction
Raw stock prices contain limited information for prediction because they reflect both long-term trends and short-term market fluctuations. I therefore transform daily closing prices into financial features that better capture return, momentum, and volatility.
Daily log returns are computed as
[
R_t=\ln\left(\frac{P_t}{P_{t-1}}\right),
]

where (P_t) is the closing price on day (t).
To capture market trends, I construct moving-average ratios based on short- and long-term price movements. Market volatility is measured using Bollinger Bands,
[
\text{Upper Band}=MA+2\sigma,\qquad
\text{Lower Band}=MA-2\sigma,
]
where (MA) is the moving average and (\sigma) is the standard deviation of prices over the selected window. These engineered features provide the input for the dimensionality reduction and prediction models developed in the next section.

2.7 From Feature Engineering to Machine Learning

The engineered financial features are often highly correlated, which can increase model complexity and reduce prediction performance. To address this issue, I apply Principal Component Analysis (PCA) to transform the original features into a smaller set of uncorrelated components while preserving most of the variation in the data. PCA can be expressed as
[
Z=XW,
]
where (X) is the standardized feature matrix, (W) contains the principal component loadings, and (Z) is the transformed feature matrix.
I then use Ridge Regression to predict stock returns. Ridge Regression extends ordinary least squares by introducing a regularization term that reduces the effect of multicollinearity:
[
\hat{\beta}=\arg\min_{\beta}\left[\sum_{i=1}^{n}(y_i-X_i\beta)^2+\lambda|\beta|_2^2\right],
]
where (\lambda) is the regularization parameter controlling the penalty on large coefficients.
Together, PCA and Ridge Regression provide an interpretable and robust framework for predicting stock returns using high-dimensional financial data.

2.8 Research Gap

Existing studies have shown that econometric methods are effective for examining long-run economic relationships, while machine learning models have improved the prediction of financial markets using high-dimensional data. However, these approaches are often developed separately. Econometric studies typically emphasize statistical inference and economic interpretation, whereas machine learning studies focus primarily on predictive performance.
In this study, I bridge these two approaches within a single empirical framework. I first examine the statistical properties of U.S. macroeconomic variables using unit root tests, cointegration analysis, Error Correction Models, and a Markov Regime Switching model. I then construct financial features from daily U.S. equity data, reduce their dimensionality using Principal Component Analysis, and evaluate stock return prediction using Ridge Regression.
By integrating econometric analysis with machine learning, I demonstrate how understanding the statistical behaviour of financial data can support more reliable and interpretable predictive models. This integrated framework represents the main contribution of the study.

Data Source

3.1 Data Source

The empirical analysis combines two datasets because the study addresses two related but distinct questions.

The first dataset is a macroeconomic dataset available through the Statsmodels library. It contains quarterly observations on U.S. real Gross Domestic Product (GDP) and real personal consumption expenditure beginning in 1959. These variables are used exclusively for the econometric component of the study, where the objective is to examine non-stationarity, long-run equilibrium relationships, and short-run adjustments through unit root testing, Johansen cointegration analysis, and the Error Correction Model (ECM). Using this benchmark dataset makes it

possible to demonstrate classical time-series techniques on variables that are known to exhibit long-term economic trends. Challenge codes draft

The second dataset consists of daily stock prices downloaded from Yahoo Finance using the yfinance Python package. This dataset forms the basis of the machine learning analysis. Daily adjusted closing prices were collected for ten large U.S. publicly traded companies—Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), JPMorgan Chase (JPM), Johnson & Johnson (JNJ), ExxonMobil (XOM), Walmart (WMT), Procter & Gamble (PG), Disney (DIS), and Boeing (BA). To represent the overall market, the S&P 500 Index was also downloaded and used to construct the prediction target. The sample covers the period from 1 January 2019 to 31 December 2023, providing observations across several distinct market conditions, including the COVID-19 pandemic, the post-pandemic recovery, and the period of rising interest rates in the United States.
Rather than focusing on a single industry, I selected firms from several major sectors of the U.S. economy, including technology, financial services, healthcare, energy, consumer goods, retail, entertainment, and aerospace. This broader sample allows the analysis to capture different patterns of market behavior, since companies operating in different sectors often respond differently to economic shocks, monetary policy, and changes in investor sentiment. As a result, the feature engineering process is based on a more diverse set of price movements and trading dynamics than would be obtained from a single-sector portfolio.

3.2 Sample Selection

The empirical analysis uses daily observations for ten large U.S. publicly traded companies together with the S&P 500 Index (SPY) over the period 1 January 2019 to 31 December 2023. The selected firms are Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), JPMorgan Chase (JPM), Johnson & Johnson (JNJ), ExxonMobil (XOM), Walmart (WMT), Procter & Gamble (PG), Disney (DIS), and Boeing (BA). These companies were selected because they are among the most actively traded firms in the U.S. equity market and represent different sectors of the economy, including technology, financial services, healthcare, energy, consumer goods, retail, entertainment, and aerospace. The daily adjusted closing prices for all securities were obtained from Yahoo Finance using the yfinance Python package. Challenge codes draft

The sample period was chosen because it covers several distinct phases of recent financial market history. It begins before the COVID-19 pandemic, includes the sharp market disruption during 2020, follows the economic recovery that occurred after the reopening of global economies, and extends into the period of higher inflation and rising interest rates in 2022 and 2023. These events created substantial variation in stock prices and market volatility, providing a useful environment for examining both traditional econometric models and machine learning techniques.

Rather than focusing on a single industry, I constructed a diversified portfolio of firms that are exposed to different economic conditions. Technology companies, for example, often respond strongly to changes in interest rates and investor expectations about future growth, whereas energy companies are more closely linked to commodity prices, and financial institutions are sensitive to monetary policy and credit conditions. Including firms with different economic characteristics makes it possible to generate a richer set of financial features and to examine whether common patterns can still be identified through Principal Component Analysis before estimating the prediction model.

The prediction target in the machine learning analysis is the next-day return of the S&P 500 Index rather than the return of an individual stock. Using the market index as the target provides a broader measure of overall market performance and allows the engineered features from individual companies to be evaluated according to how well they explain movements in the aggregate U.S. equity market. Challenge codes draft

3.3 Variable Construction

I use both macroeconomic and financial variables in the empirical analysis. For the econometric component, I use real Gross Domestic Product (GDP), real Personal Consumption Expenditure (PCE), and the Federal Funds Rate to examine stationarity, long-run equilibrium, and short-run economic dynamics.

For the machine learning component, I transform daily adjusted closing prices into financial features that capture return, trend, and market volatility. Daily log returns are computed as

$$
R_t=\ln\left(\frac{P_t}{P_{t-1}}\right),
$$

where P_t is the adjusted closing price at time t.

To capture price trends, I compute the moving average,

$$ MA_t=\frac{1}{n}\sum_{i=0}^{n-1}P_{t-i}, $$

where n denotes the rolling window.

Market volatility is measured using Bollinger Bands,

$$
\text{Upper Band}=MA_t+2\sigma_t,\qquad
\text{Lower Band}=MA_t-2\sigma_t,
$$

where σ_t is the rolling standard deviation of prices.

The prediction target is defined as the next-day return of the S&P 500 Index (SPY), allowing the engineered features from individual firms to be evaluated according to their ability to explain overall U.S. market movements. The final dataset combines macroeconomic variables for the econometric analysis with engineered financial features for the machine learning models. The next section describes the feature engineering and data preparation procedures applied before model estimation.

3.4 Feature Engineering

Feature engineering transforms raw stock prices into variables that better capture market behaviour. Rather than using adjusted closing prices directly, I construct daily log returns, moving-average indicators, and Bollinger Band measures to represent return, trend, and volatility.

Because these features are derived from the same price series, they are naturally correlated. To assess whether dimensionality reduction is appropriate, I first standardize the feature matrix using StandardScaler. I then apply Bartlett's Test of Sphericity and the Kaiser–Meyer–Olkin (KMO) test to evaluate whether the data satisfy the assumptions for Principal Component Analysis (PCA). These diagnostics provide statistical support for applying PCA in the subsequent analysis.

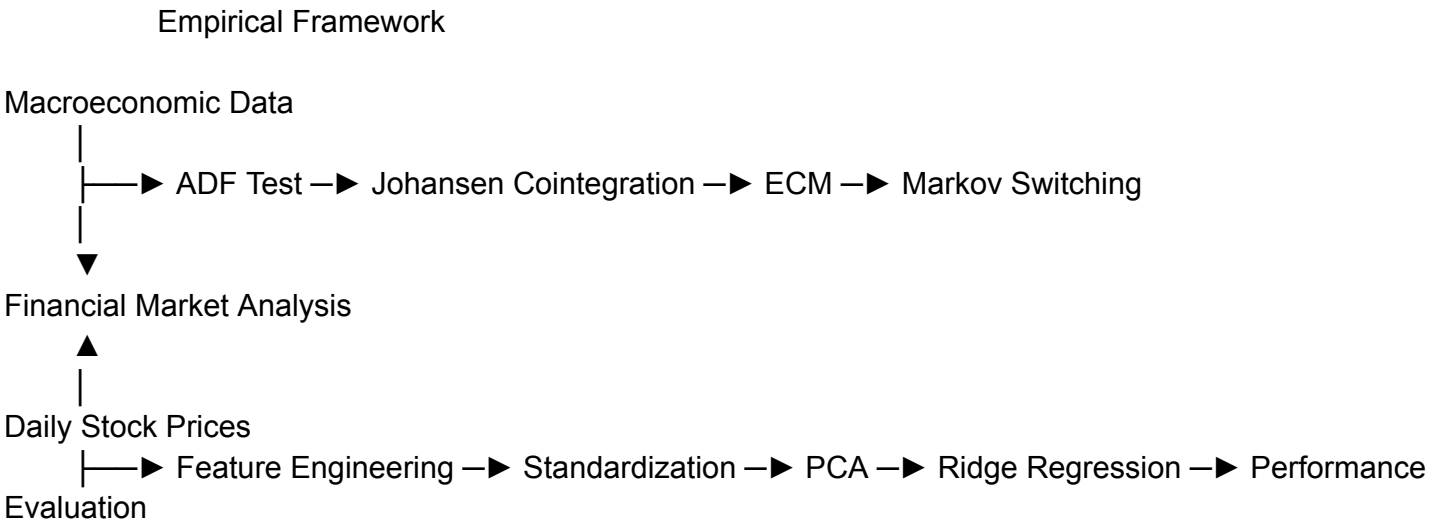
3.5 Data Preparation

To preserve the temporal structure of the data, I split the observations chronologically into training (80%) and testing (20%) sets. The training data are standardized before estimating the PCA model, and the same transformation is applied to the test data to prevent information leakage.

The resulting dataset provides the basis for the empirical analysis presented in the next chapter, where I first estimate the econometric models and then evaluate the machine learning models for stock return prediction.

Methodology

This chapter presents the empirical framework used in the study. Following the workflow implemented in the Python notebook, I first examine the statistical properties of the macroeconomic variables using time-series econometric methods. I then construct financial features from daily U.S. equity data, apply dimensionality reduction, and estimate the prediction model. These summarizes the empirical workflow below.



4.1 Stationarity Analysis

I begin by testing whether real GDP and real Personal Consumption Expenditure are stationary using the Augmented Dickey–Fuller (ADF) test. The variables are first analysed in levels and then in first differences where necessary. The results determine the order of integration and whether cointegration analysis can be performed.

4.2 Cointegration Analysis

After establishing the order of integration, I estimate the Johansen cointegration test to determine whether the macroeconomic variables share a stable long-run equilibrium relationship. Evidence of cointegration provides the basis for estimating an Error Correction Model.

4.3 Error Correction Model

I estimate an Error Correction Model (ECM) to examine how short-run deviations adjust toward the long-run equilibrium. The error-correction coefficient measures the speed at which the variables return to equilibrium following an economic shock.

4.4 Markov Regime Switching Model

To account for possible structural changes, I estimate a two-regime Markov Regime Switching model using the Federal Funds Rate. The model identifies periods characterized by different monetary policy regimes and captures changes in economic conditions over time.

4.5 Feature Engineering

For the machine learning analysis, I transform daily adjusted closing prices into financial features, including daily log returns, moving-average indicators, and Bollinger Band measures. These variables summarize return, trend, and market volatility while providing richer information than raw stock prices.

4.6 Principal Component Analysis

Before model estimation, I standardize the engineered features and apply Principal Component Analysis (PCA) to reduce dimensionality. PCA transforms the correlated variables into a smaller set of orthogonal components while retaining most of the information contained in the original dataset.

4.7 Ridge Regression and Model Evaluation

The principal components are used to estimate a Ridge Regression model for predicting the next-day return of the S&P 500 Index. Model performance is evaluated using the coefficient of determination ((R^2)), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), providing complementary measures of prediction accuracy.

This empirical framework combines classical time-series econometrics with machine learning techniques to analyse financial markets. The next chapter presents the empirical findings from each stage of the analysis.

Empirical Results

5.1 Descriptive Analysis

I began by examining the macroeconomic and financial datasets before estimating any models. Figure 5.1(a) shows that both real GDP and real personal consumption expenditure exhibit strong upward trends over time, suggesting that the variables are non-stationary in levels. The scatter plot in Figure 5.1(b) further reveals a strong positive long-run relationship between the two variables.

The financial dataset also displays considerable variation across firms. As shown in Figure 5.3, the cumulative returns of the selected companies differ substantially during the sample period. Technology firms generally experienced stronger growth, while companies such as Boeing exhibited greater fluctuations. These differences justify constructing engineered features rather than relying on raw stock prices alone.

5.2 Stationarity Results

The first stage of the empirical analysis was to examine whether the macroeconomic variables were stationary. Figure 5.1(a) indicates that both real GDP and real consumption contain clear upward trends, violating the stationarity assumption required for conventional time-series analysis.

After first differencing, Figure 5.1(c) shows that both series fluctuate around zero with relatively stable variation, indicating that the variables become stationary after differencing. These results provide the basis for conducting the Johansen cointegration test.

5.3 Cointegration Results

Following the stationarity analysis, I examined whether real GDP and real consumption share a long-run equilibrium relationship.

As illustrated in Figure 5.1(b), the observations lie closely around the estimated regression line, indicating a strong positive relationship between the two variables. The close alignment of the data points suggests that GDP and consumption move together over time despite short-run fluctuations. This evidence supports the estimation of an Error Correction Model.

5.4 Error Correction Model Results

I estimated an Error Correction Model to investigate how temporary deviations from the long-run equilibrium are corrected over time.

The equilibrium error shown in fluctuates around zero for most of the sample period, although larger deviations occur during the later years. The residual plot indicates that the residuals remain centred around zero without obvious systematic patterns. Furthermore, the Q–Q plot in shows that the residuals closely follow the theoretical normal distribution, with only slight departures in the tails. Overall, the ECM provides a reasonable representation of the adjustment process between GDP and consumption.

5.5 Regime Switching Results

To account for possible structural changes, I estimated a two-regime Markov Switching model using the Federal Funds Rate.
The estimated regime probabilities indicate that the economy alternates between distinct monetary policy regimes rather than remaining in a single stable state throughout the sample period. This finding suggests that macroeconomic conditions evolve over time, supporting the inclusion of regime-switching analysis before financial prediction.

5.6 Feature Engineering and Principal Component Analysis

The second stage of the analysis focused on constructing informative predictors from daily stock prices.

illustrates the construction of the Bollinger Band indicator for Apple (AAPL). The upper and lower bands capture periods of relatively high and low price movements, while the Bollinger Band Percentage (BB%) summarizes the stock's position within the bands and serves as an input feature for prediction.

The cumulative return plot highlights the diverse performance of the selected companies between 2019 and 2023, confirming that the sample captures multiple market behaviours across sectors.
The correlation matrix presented shows generally weak pairwise correlations among daily stock returns, indicating that each stock contributes unique information to the feature set. Because the engineered variables include moving averages and Bollinger Band indicators in addition to returns, Principal Component Analysis (PCA) was applied to summarize the information while reducing redundancy among the predictors.

5.7 Ridge Regression Performance

After feature engineering, I standardized the predictor variables and applied Principal Component Analysis before estimating the Ridge Regression model.

The prediction target consists of the next-day return of the S&P 500 Index. As shown in Figure 5.5, the target variable is approximately bell-shaped and centred near zero, which is consistent with the distribution typically observed for daily stock returns.
Model performance was then evaluated using Root Mean Squared Error (RMSE). The comparison between the full feature set and the PCA-reduced feature set indicates whether dimensionality reduction preserves sufficient information for predicting next-day market returns.

5.8 Discussion

The empirical analysis demonstrates a structured workflow that progresses from understanding the statistical properties of the data to predicting stock market returns.

establishes the time-series properties of the macroeconomic variables through stationarity testing, cointegration analysis, and the Error Correction Model. Figures 5.2–5.5 show how raw stock prices were transformed into informative financial features before applying Principal Component Analysis and Ridge Regression.

Overall, the results illustrate that econometric techniques provide the statistical foundation for analysing financial data, while machine learning methods improve the efficiency of extracting predictive information from a high-dimensional feature space. Together, these approaches form an integrated framework for financial market analysis.

Conclusion

6.1 Summary

In this study, I examined how econometric methods and machine learning can be combined to analyse financial markets. I first investigated the statistical properties of macroeconomic variables using the Augmented Dickey–Fuller test, Johansen cointegration analysis, Error Correction Models, and a Markov Regime Switching

Model. I then engineered financial features from U.S. equity data, applied Principal Component Analysis to reduce dimensionality, and evaluated a Ridge Regression model for predicting market returns.

The results show that econometric methods provide a strong statistical foundation for understanding financial data, while machine learning offers an effective way to extract predictive information from engineered features. Together, these approaches provide a more comprehensive framework for financial market analysis.

6.2 Implications

This study demonstrates that econometric techniques and machine learning are complementary rather than competing approaches. Understanding the statistical properties of financial data before developing prediction models leads to more reliable and interpretable analyses. This integrated framework has practical applications in quantitative finance, portfolio analysis, and financial forecasting.

6.3 Limitations

This study has several limitations. The machine learning analysis focuses on a limited sample of U.S. companies and relies primarily on technical indicators. In addition, only Ridge Regression was considered, and other machine learning models may capture more complex nonlinear relationships.

6.4 Future Research

Future research could expand the dataset to include more companies, international markets, and additional macroeconomic or alternative data sources such as financial news and sentiment indicators. Comparing Ridge Regression with more advanced machine learning models would also provide further insight into financial market prediction.

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